

YANG FU

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EDUCATION

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- Ph.D. in Electrical and Computer Engineering**, University of California San Diego March, 2026
 Thesis: From 3D Reconstruction to Spatial Intelligence: Bridging Vision Foundation Models with the 3D World
 Advisor: **Prof. Xiaolong Wang**
- M.S. in Electrical and Computer Engineering**, University of Illinois, Urbana-Champaign May, 2021
 Thesis: Towards Accurate Person Re-Identification by Deep Learning 📄
 Advisor: **Prof. Thomas S. Huang** and **Prof. Humphrey Shi**
- B.S. in Electrical and Computer Engineering**, Beihang University June, 2017
 Thesis: Detection of optic nerve sheath pulsation in ultrasound video

CONFERENCE PUBLICATIONS AND PREPRINTS

-
1. Yatai Ji, An-Chieh Cheng, **Yang Fu**, Yukang Chen, Han Zhang, Zhaojing Yang, Wei Huang, Ka Chun Cheung, Song Han, Vidya Nariyambut Murali, Pavlo Molchanov, Jan Kautz, Hongxu Yin, Ping Luo, Sifei Liu Reinforcing Spatial Reasoning Paths in Vision Language Models *Under review*.
 2. Yingwei Li, Xin Huang, Yang Liu, **Yang Fu**, Alex Zihao Zhu, Chen Song, Junwen Yao, Anant Subramanian, Hao Xiang, Weijing Shi, Yuliang Zou, Zhaoqi Leng, Govind Thattai, Dragomir Anguelov, Mingxing Tan. STELLAR: Scaling 3D Perception Large Models for Autonomous Driving. *Under review*.
 3. **Yang Fu**, Sifei Liu, Hongxu Yin, Benjamin Eckar, Jan Kautz, Xiaolong Wang, Arash Vahdat, Chao Liu. Bridging 2D Vision Language Models to 3D World via Feature Distillation. *Under review*.
 4. **Yang Fu**, Yuliang Zou, Hao Xiang, Xin Huang, Yijing Bai, Chen Song, Weijing Shi, Govind Thattai, Dragomir Anguelov, Mingxing Tan, Yingwei Li. Scene Reconstruction as Mapping Priors for 3D Detection. **Highlight presentation** at *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
 5. An-Chieh Cheng, **Yang Fu**, Yatai Ji, Ligeng Zhu, Guanqi Zhan, Zhuoyang Zhang, Zhaojing Yang, Song Han, Yao Lu, Pavlo Molchanov, Vidya Nariyambut Murali, Jan Kautz, Xiaolong Wang, Hongxu Yin, Sifei Liu. Grounded 3D-Aware Spatial Vision-Language Modeling. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
 6. An-Chieh Cheng, **Yang Fu**, Yukang Chen, Zhijian Liu, Xiaolong Li, Subhashree Radhakrishnan, Jan Kautz, Song Han, Yao Lu, Hongxu Yin, Xiaolong Wang, Sifei Liu. 3D Aware Region Prompted Vision Language Model. *International Conference on Learning Representations*, 2026.
 7. Yutao Xie, Nathaniel Thomas, Nicklas Hansen, **Yang Fu**, Li Erran Li, Xiaolong Wang. TIPS: Turn-level Information-Potential Reward Shaping for Search-Augmented LLMs. *International Conference on Learning Representations*, 2026.
 8. Ri-Zhao Qiu*, Yafei Hu*, Ge Yang, Yuchen Song, **Yang Fu**, Jianglong Ye, Jiteng Mu, Ruihan Yang, Nikolay Atanasov, Sebastian Scherer, Xiaolong Wang. Learning Generalizable Feature Fields for Mobile Manipulation. Arxiv, 2024.
 9. An-Chieh Cheng, Hongxu Yin, **Yang Fu**, Qiushan Guo, Ruihan Yang, Jan Kautz, Xiaolong Wang, Sifei Liu. SpatialRGPT: Grounded Spatial Reasoning in Vision Language Model. *Conference on Neural Information Processing Systems*, 2024.
 10. Kaiwen Jiang, **Yang Fu**, Mukund Varma T, Yash Belhe, Xiaolong Wang, Hao Su, Ravi Ramamoorthi. A Construct-Optimize Approach to Sparse View Synthesis without Camera Pose. *ACM SIGGRAPH*, 2024.

11. **Yang Fu**, Sifei Liu, Amey Kulkarni, Jan Kautz, Alexei A. Efros, Xiaolong Wang. COLMAP-Free 3D Gaussian Splatting. **Highlight presentation** at *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
12. Hongchi Xia*, **Yang Fu***, Sifei Liu, Xiaolong Wang. RGBD Objects in the Wild: Scaling Real-World 3D Object Learning from RGB-D Videos. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. (* equal contribution)
13. Mengqi Zhang*, **Yang Fu***, Zheng Ding, Sifei Liu, Zhuowen Tu, Xiaolong Wang. HOIDiffusion: Generating Realistic 3D Hand-Object Interaction Data. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. (* equal contribution)
14. **Yang Fu**, Shalini De Mello, Xueting Li, Amey Kulkarni, Jan Kautz, Xiaolong Wang, Sifei Liu. 3D Reconstruction with Generalizable Neural Fields using Scene Priors. *International Conference on Learning Representations*, 2024.
15. **Yang Fu**, Ishan Misra, Xiaolong Wang. MonoNeRF: Learning Generalizable NeRFs from Monocular Videos without Camera Poses. *International Conference on Machine Learning*, 2023.
16. Kaifeng Zhang, **Yang Fu**, Shubhankar Borse, Hong Cai, Fatih Porikli, Xiaolong Wang. Self-supervised Geometric Correspondence for Category-level 6D Object Pose Estimation in the Wild. *International Conference on Learning Representations*, 2023.
17. **Yang Fu**, Xiaolong Wang. Category-Level 6D Object Pose Estimation in the Wild: A Semi-Supervised Learning Approach and A New Dataset. *Conference on Neural Information Processing Systems*, 2022.
18. Yuzhe Qin, Yueh-Hua Wu, Shaowei Liu, Hanwen Jiang, Ruihan Yang, **Yang Fu**, Xiaolong Wang. DexMV: Imitation Learning for Dexterous Manipulation from Human Videos. *European Conference on Computer Vision*, 2022.
19. **Yang Fu**, Sifei Liu, Shalini De Mello, Umar Iqbal, Humphrey Shi, Jan Kautz. Learning to Track Instances without Video Annotations. **Oral presentation** at *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021.
20. **Yang Fu**, Linjie Yang, Ding Liu, Thomas Huang, Humphrey Shi. Comprehensive Feature Aggregation for Video Instance Segmentation. *AAAI Conference on Artificial Intelligence*, 2021.
21. **Yang Fu**, Yunchao Wei, Guanshuo Wang, Yuqian Zhou, Honghui Shi, Thomas Huang. Self-similarity Grouping: A Simple Unsupervised Cross-Domain Adaptation Approach for Person Re-Identification. **Oral presentation** at *IEEE/CVF International Conference on Computer Vision*, 2019.
22. **Yang Fu**, Xiaoyang Wang, Yunchao Wei, Thomas S.Huang. STA: Spatial-Temporal Attention for Large-Scale Video-based Person Re-Identification. *AAAI Conference on Artificial Intelligence*, 2019.
23. **Yang Fu**, Yunchao Wei, Yuqian Zhou, Honghui Shi, Gao Huang, Xinchao Wang, Thomas.S Huang. Horizontal Pyramid Matching for Person Re-identification. *AAAI Conference on Artificial Intelligence*, 2019.

HONORS AND FELLOWSHIPS

Qualcomm Innovation Fellowship	2023
UCSD Graduate Fellowship	2021

WORKING EXPERIENCE

Research Intern , Waymo, USA	June, 2025 - Jan, 2026
Research Assistant , UC San Diego, USA	Sep, 2021 - present
Research Intern (Part time) , Nvidia Research, USA	Sep, 2024 - Jun, 2025
Research Intern , Nvidia Research, USA	Jun, 2024 - Sep, 2024
Research Intern (Part time) , Nvidia Research, USA	Sep, 2023 - Jun, 2024
Research Intern (Part time) , Nvidia Research, USA	Sep, 2022 - March, 2023

Research Intern, Nvidia Research, USA

June, 2022 - Sep, 2022

Research Intern, Nvidia Research, USA

May, 2020 - Dec, 2020

Research Intern, TikTok AI Lab, USA

May, 2019 - Aug, 2019

Research Intern, Nokia Bell Labs, USA

Jun, 2018 - Aug, 2018

TEACHING EXPERIENCE

Teaching Assistant

San Diego, C.A.

- ECE 285
- ECE 176

March, 2023 - Jun, 2023

Jan, 2024 - March, 2024 & Jan, 2025 - March, 2025

SERVICE

Manuscript Reviewer:

TIP, T-PAMI, CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, MICCAI

Organizing Committee:

Workshop on Large-scale Video Object Segmentation Challenge, *CVPR* 2022

Workshop on Large-scale Video Object Segmentation Challenge, *CVPR* 2021.